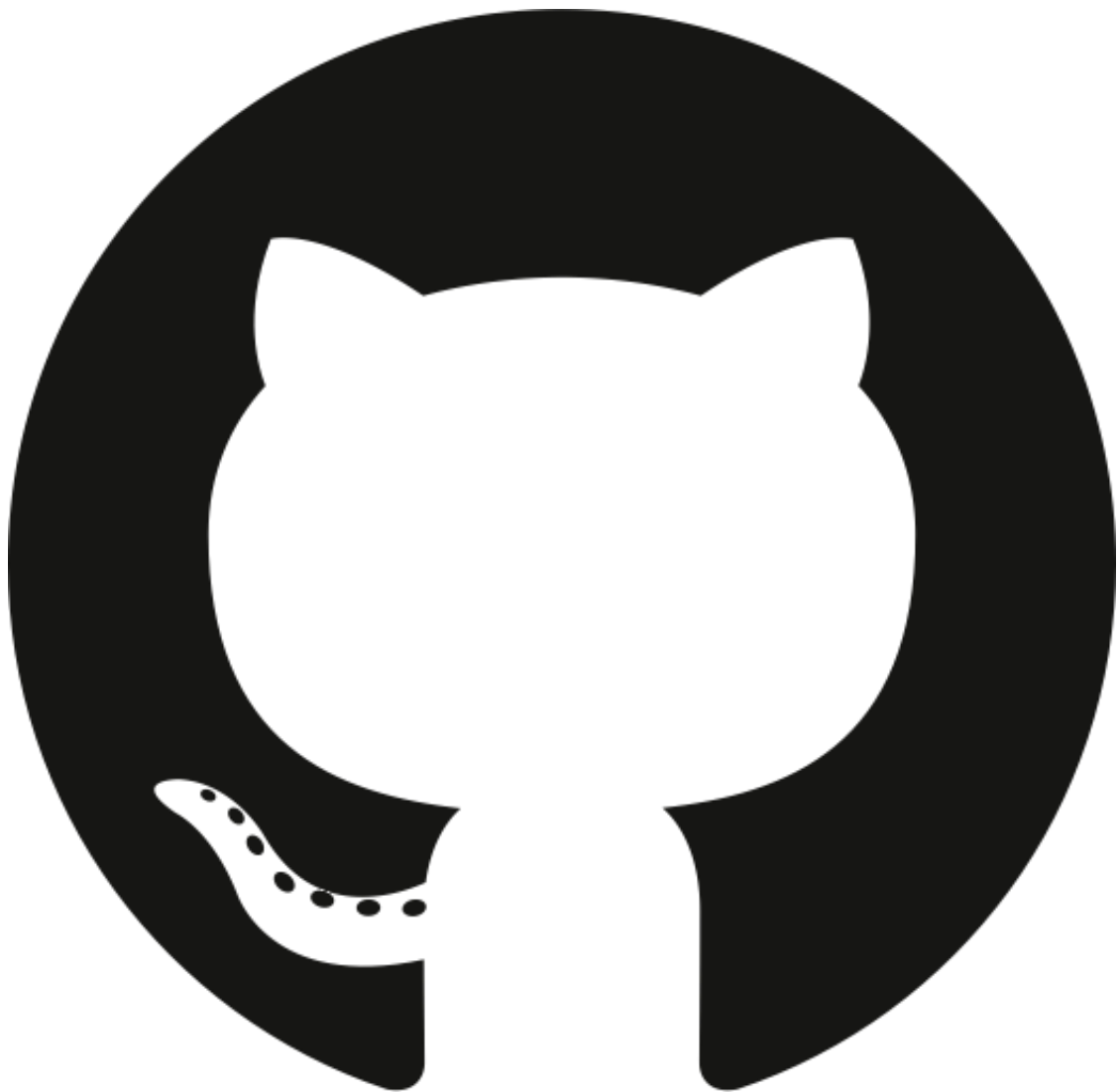


GITHUB COMMANDS CHEATSHEET

(A COMPLETE QUICK-REFERENCE GUIDE)



1. Git Configuration

- `git config --global user.name "Your Name"` : Sets your global Git username.
- `git config --global user.email "you@example.com"` : Sets your global Git email address.
- `git config --list` : Displays all Git configuration settings.
- `git config --global core.editor "code --wait"` : Sets VS Code as the default Git editor.

2. Creating & Cloning Repository

- `git init` : Initializes a new local Git repository.
- `git clone <repository_url>` : Clones a remote repository to your local machine.
- `git clone <repository_url> <folder_name>` : Clones repository into a specified folder.

3. Basic Snapshotting

- `git add <file>` : Stages a specific file for commit.
- `git add .` : Stages all files in the current directory for commit.
- `git status` : Displays the state of the working directory and staging area.
- `git diff` : Shows changes between working directory and staging area.
- `git commit -m "commit message"` : Commits staged changes with a message.
- `git commit -am "commit message"` : Adds and commits tracked files in one step.

4. Branching & Merging

- `git branch` : Lists all branches in the repository.
- `git branch <branch_name>` : Creates a new branch.
- `git checkout <branch_name>` : Switches to the specified branch.
- `git switch <branch_name>` : Alternative to checkout for switching branches.
- `git merge <branch_name>` : Merges the specified branch into the current branch.
- `git branch -d <branch_name>` : Deletes a branch safely.
- `git branch -D <branch_name>` : Force deletes a branch.

5. Sharing & Updating Projects

- `git remote add origin <repository_url>` : Adds a new remote repository connection.
- `git remote -v` : Displays remote repositories linked to your project.
- `git fetch` : Downloads objects and refs from another repository.
- `git pull origin main` : Fetches and merges changes from the main branch.
- `git push origin main` : Uploads local commits to the main branch on the remote.
- `git push -u origin <branch_name>` : Sets the upstream branch for future pushes.

- `git remote remove origin` : Removes the linked remote repository.

6. Stashing Changes

- `git stash` : Temporarily saves uncommitted changes for later use.
- `git stash list` : Displays all stashed changes.
- `git stash apply` : Applies the most recent stash without deleting it.
- `git stash pop` : Applies and removes the most recent stash.
- `git stash drop` : Deletes a specific stash entry.
- `git stash clear` : Removes all stashed entries.

7. Viewing History

- `git log` : Displays the full commit history.
- `git log --oneline` : Shows compact commit history.
- `git log --graph --oneline --decorate --all` : Visualizes branch history in a graph format.
- `git show <commit_id>` : Displays details about a specific commit.
- `git diff <commit_id> <commit_id>` : Compares changes between two commits.

8. Undoing Changes

- `git checkout -- <file>` : Restores file from the last commit.
- `git revert <commit_id>` : Creates a new commit that undoes the specified commit.
- `git reset --soft <commit_id>` : Resets index to a specific commit, keeping changes staged.
- `git reset --hard <commit_id>` : Completely resets to a specific commit, discarding changes.
- `git clean -fd` : Removes all untracked files and directories.

9. GitHub Specific Commands

- `git remote add origin https://github.com/username/repo.git` : Links local repo to GitHub.
- `git push -u origin main` : Pushes changes and sets upstream for the main branch.
- `git pull origin main` : Pulls the latest changes from GitHub.
- `git clone https://github.com/username/repo.git` : Clones a GitHub repository.
- `git fetch --all --prune` : Fetches all branches and removes deleted ones.
- `git remote set-url origin https://github.com/new/repo.git` : Updates remote repository URL.

10. Tagging

- `git tag <tag_name>` : Creates a lightweight tag.
- `git tag -a <tag_name> -m "message"` : Creates an annotated tag with a message.
- `git push origin <tag_name>` : Pushes tag to remote repository.

- `git tag -d <tag_name>` : Deletes a local tag.
- `git push origin --delete <tag_name>` : Deletes a tag from remote repository.
- `git tag` : Lists all tags.

11. Collaboration Commands

- `git fetch origin` : Fetches changes from remote repository.
- `git merge origin/main` : Merges latest main branch updates locally.
- `git rebase origin/main` : Reapplies commits on top of the latest main branch.
- `git push origin HEAD` : Pushes the current branch to remote repository.
- `git cherry-pick <commit_id>` : Applies a specific commit from another branch.

12. Advanced Commands

- `git reflog` : Shows reference history for all HEAD movements.
- `git bisect start` : Starts a binary search to find buggy commit.
- `git bisect bad` : Marks current commit as bad during bisect.
- `git bisect good <commit_id>` : Marks a commit as good during bisect.
- `git blame <file>` : Shows who changed which line in a file.
- `git shortlog -sn` : Shows number of commits by each author.

13. GitHub SSH & Authentication

- `ssh-keygen -t rsa -b 4096 -C "your_email@example.com"` : Generates a new SSH key.
- `eval "$(ssh-agent -s)"` : Starts the SSH agent in background.
- `ssh-add ~/.ssh/id_rsa` : Adds SSH key to the agent.
- `cat ~/.ssh/id_rsa.pub` : Displays your public SSH key.
- `git remote set-url origin git@github.com:username/repo.git` : Sets SSH URL for your remote repo.

14. Submodules

- `git submodule add <repository_url>` : Adds a Git submodule.
- `git submodule update --init --recursive` : Initializes and updates submodules.
- `git submodule foreach git pull origin main` : Pulls latest changes for all submodules.
- `git submodule deinit -f .` : Deinitializes submodules.

15. Cleaning Up & Maintenance

- `git gc` : Cleans up unnecessary files and optimizes repository.
- `git prune` : Removes unreachable Git objects.
- `git fsck` : Verifies integrity of Git objects.
- `git repack` : Repacks Git objects for efficiency.